

Instructions: Use proper notation, and show all of your work to receive credit. If you are doing a proof, justify every step.

HANDOUT 2 → 1. (1 pt) Let A be some arbitrary event and S be the sample space. Which of the following is not one of the three axioms of probability?
(pg. 2)

- (a) $P(S) = 1$
(b) $P(A) \geq 0$
C. (c) $P(A) = \frac{n_A}{N}$, where n_A is the number of sample points in A and N is the number of sample points in S
(d) If $A_1, A_2, \dots$ are pairwise mutually disjoint, i.e., $A_i \cap A_j = \emptyset$ for $i \neq j$, then $P(\bigcup_i A_i) = \sum_i P(A_i)$

HANDOUT 4 → 2. (2 pts) In the lecture notes, we gave two ways to write $P(A \cap B)$ using the Multiplicative Law. Provide one of them.
(pg. 2)

$$P(A \cap B) = P(A) \cdot P(B|A)$$

OR

$$P(A \cap B) = P(B) \cdot P(A|B)$$

HANDOUT 1 → 3. (1 pt) True or False (circle your answer): If A is a subset of B (i.e., $A \subseteq B$), then it is possible that $A = B$.
(pg. 2)

True

False

HANDOUT 1 → 4. (2 pts) One of DeMorgan's Laws says that $\overline{A \cap B} = \overline{A} \cup \overline{B}$. What does the other one say?
(pg. 3)

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

Section 2.7 → 5. (3 pts) Suppose that A and B are mutually exclusive events, with $P(A) > 0$ and $P(B) > 0$. Are A and B independent? Prove your answer.
(problem 79)

Suppose A and B are mutually exclusive with $P(A) > 0$ and $P(B) > 0$.

Since A and B are mutually exclusive, $P(A \cap B) = 0$.

However, since $P(A) > 0$ and $P(B) > 0$, $P(A)P(B) > 0$.

Thus, $P(A \cap B) \neq P(A)P(B)$.

Hence, A and B are not independent. $\square$

Section 2.6 → 6. (3 pts) An airline has six flights from New York to California and seven flights from California to Hawaii per day. If the flights are to be made on separate days, how many different flight arrangements can the airline offer from New York to Hawaii?
(problem 35)

By the Fundamental Counting Principle, there are
 $(6)(7) = 42$ different flight arrangements from
New York to Hawaii.

Section 2.6 →
(problem 54)

7. (3 pts) A group of three undergraduate and five graduate students are available to fill certain student government posts. If four students are to be randomly selected from this group, find the probability that exactly two undergraduates will be among the four chosen.

$$\begin{aligned} P(\text{exactly 2 undergraduates selected}) &= \frac{\binom{3}{2} \binom{5}{2}}{\binom{8}{4}} \\ &= \frac{30}{70} \\ &\approx .4286 \end{aligned}$$

Section 2.7 →
(problem 71)

8. (3 pts) If two events, A and B , are such that $P(A) = .5$, $P(B) = .3$, and $P(A \cap B) = .1$, find $P(A|B)$.

$$\begin{aligned} P(A|B) &= \frac{P(A \cap B)}{P(B)} \quad (\text{by definition of conditional probability}) \\ &= \frac{.1}{.3} \\ &= .3333 \quad \text{OR} \quad 1/3 \end{aligned}$$

Section 2.8 →
(problem 94)

9. (3 pts) A smoke detector system uses two devices, A and B . If smoke is present, the probability that it will be detected by device A is .95; by device B , .90; and by both devices, .88. If smoke is present, find the probability that the smoke will be detected by either device A or B or both devices.

Given: $P(A) = .95$, $P(B) = .9$, $P(A \cap B) = .88$

Want: $P(A \cup B)$

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \quad (\text{by Additive Law}) \\ &= .95 + .9 - .88 \\ &= .97 \end{aligned}$$

- Section 2.8 →
(problem 96)
10. (4 pts) If A and B are independent events with $P(A) = .5$ and $P(B) = .2$, find $P(\overline{A} \cup \overline{B})$.

$$\begin{aligned} P(\overline{A} \cup \overline{B}) &= P(\overline{A \cap B}) = 1 - P(A \cap B) = 1 - P(A)P(B) = 1 - (.5)(.2) = .9 \end{aligned}$$

$\uparrow$ $\uparrow$ $\uparrow$
DeMorgan's Law Complementary Law independence